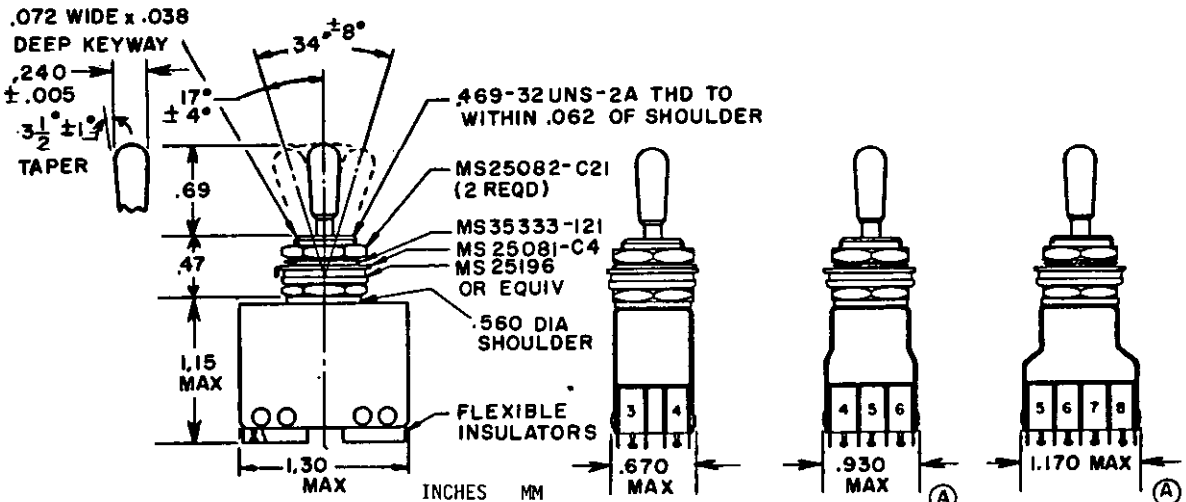
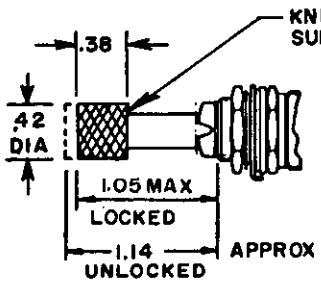
MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, SENSITIVE, TOGGLE ACTUATED,
7 AMPERES, PANEL SEALED, RFI GROUNDED

This specification sheet is approved for use by the Engineering and Technical Support Division (AFLC/LOIE) Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the switch assemblies described herein shall consist of this document and the latest issue of Specification MIL-S-8805.





LEVER-LOCK TOGGLE

INCHES	MM
.005	.13
.038	.97
.050	1.27
.062	1.57
.072	1.83
.240	6.10
.38	9.7
.42	10.7
.469	11.91
.47	11.9
.560	14.22
.670	17.02
.69	17.5
.930	23.62
1.05	26.7
1.14	29.0
1.15	29.2
1.30	33.0
1.170	29.72

FIGURE 1A (4-POLE) FIGURE 1B (6-POLE) FIGURE 1C (8-POLE) (A)

BASIC SWITCH AND TERMINAL IDENTIFICATION

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.010 (.25 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
4. Design configuration optional provided maximum dimensions are not exceeded.
5. Basic switch and terminal identification shall be permanently marked on the basic switch as shown.
6. Basic switch terminals shall be single turret (as shown) and shall accept AN-20 wire.

FIGURE 1. Configuration and dimensions.

(A) denotes changes.

FSC 5930

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REQUIREMENTS:

Dimensions and configuration:	See figure 1.
Enclosure design:	1 *(unsealed). Switches shall have a molded silicone* rubber toggle to bushing seal.
Temperature characteristic:	1 (-55°C to +85°C).
(A) Shock:	50g, test condition G, method 213 of MIL-STD-202.
Vibration:	In accordance with method 214 of MIL-STD-202, and as follows: 20-300 Hz, 3 dB/octave increase 300-1,000 Hz, 0.02 G^2/Hz 1,000-2,000 Hz, 6 dB/octave decrease
Toggle lever and mounting bushing materials:	Shall be stainless steel or nickel plated.
Terminals:	Terminals shall be the single turret type and may be gold plated .000050 inch maximum thickness.
Toggle lever:	Shall have a lusterless finish and shall have a fixed pivot.
Weight:	4 pole: 1.8 ounces maximum. 6 pole: 1.9 ounces maximum. 8 pole: 2.0 ounces maximum.
Dielectric withstanding voltage:	
Sea level:	1,000 Vrms.
Altitude:	400 Vrms at 70,000 feet.
	After electrical endurance, the dielectric withstanding voltage points of application between all unconnected terminals of the same pole is not applicable.
Operating characteristics:	
Locking arrangement (see figure 2):	The locking combinations for the lever-lock toggle lever shall provide positive locking and shall not permit a change in toggle lever position until the locking mechanism is manually released. When locking combinations are required, the lever-lock toggle shall be furnished with the required locking combination.
Terminal strength:	5 pounds.
Strength of actuating means:	25 pounds.
Strength of mounting bushing:	25 pound-inches shall be applied to the mounting nut.
Mechanical endurance:	50,000 cycles.
Lever-lock mechanical endurance:	Lever-lock switches shall be tested for 50,000 cycles by pulling the lever to its fully extended position and returning to its fully locked position without causing contact transfer at a rate of 60 cycles per minute maximum.
Electrical endurance:	25,000 cycles.
Toggle to bushing resistance:	100 milliohms maximum. Switches shall be tested in accordance with the following: (a) Points of measurement - Between lever and bushing with contact made to uncleaned or as received surfaces. (b) Test current - 100 ±10 milliamperes. (c) Open-circuit test voltage - 6 ±1 Vdc. (d) Number of tests - Once with the lever in each position.
Electrical ratings:	See table I.

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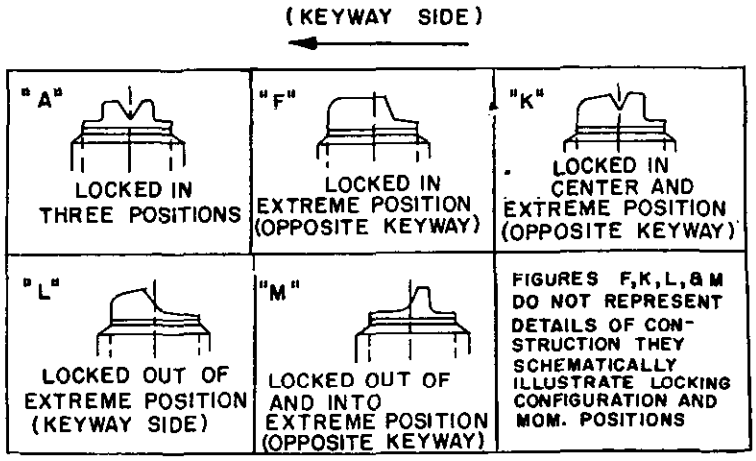


FIGURE 2. Locking arrangements.

(A) TABLE I. Electrical ratings.

Load	Silver contacts			Gold contacts
	Sea level		70,000 ft	Sea level and 70,000 ft
	28 Vdc	115 Vac, 60 Hz	28 Vdc	28 Vdc
	Amperes	Amperes	Amperes	Ampere
Resistive - - - -	7.0	7.0	3.0	1.0
Inductive - - - -	4.0	7.0	2.0	.5
Lamp - - - - - -	2.5	2.0	---	---

Part number: M8805/103- (dash number from table II).

(A) QUALITY ASSURANCE:

The following tables for qualification and group A inspections shall apply only when the basic sensitive switch is listed on QPL-8805 under Specification Sheet MIL-S-8805/4.

Qualification inspection: Per table IV.
Group A inspection: Per table V, provided in-process dielectric withstanding voltage is performed.

(A) TABLE II. Dash numbers and characteristics.

Dash number			Type number (table III)	Locking combination (see figure 2)	Contact material
Figure 1A (4-Pole)	Figure 1B (6-Pole)	Figure 1C (8-Pole)			
-001	---	-002	1	None	Silver
---	-004	---	3	None	↓
-005	---	---	5	None	↓
-009	---	---	5	M	↓
---	-010	---	3	F	↓
---	---	-011	1	A	↓
-012	---	-013	1	None	Gold
-014	---	-015	5	None	↓
-016	---	---	5	K	↓
-017	---	---	1	M	↓
---	---	-018	5	L	↓

NOTE: Dash numbers -003, -006, -007, and -008 previously covered by MIL-S-8805/103(USAF) have been canceled.

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TABLE III. Toggle action and circuitry.

Type number	Basic switch numbers			With toggle lever in position					
				Keyway side		Center		Opposite keyway side	
	Figure 1A	Figure 1B	Figure 1C	Toggle action	Circuit closed	Toggle action	Circuit closed	Toggle action	Circuit closed
1	1-2 3-4	1-2-3 4-5-6	1-2-3-4 5-6-7-8	Maintain	C-NC C-NO	Maintain	C-NC C-NC	Maintain	C-NO C-NC
3	1-2 3-4	1-2-3 4-5-6	1-2-3-4 5-6-7-8	Maintain	C-NC C-NO	(None)		Maintain	C-NO C-NC
5	1-2 3-4	1-2-3 4-5-6	1-2-3-4 5-6-7-8	Momentary	C-NC C-NO	Maintain	C-NC C-NC	Maintain	C-NO C-NC

(A) TABLE IV. Qualification group submission inspection.

Examination or test	Test sample	Other switch samples	Extent of approval
Group I Visual and mechanical examination Toggle to bushing resistance Dielectric withstanding voltage Insulation resistance Operating characteristics	Group I M8805/103-013 (8 units) M8805/103-015 (8 units)	M8805/103-012 (2 units) 1. Visual and mechanical examination 2. Operating characteristics 3. Lever-lock mechanical endurance	All part numbers
Group II Strength of actuating means 1/ Strength of mounting bushing 1/ Thermal shock Vibration (random) Shock (50g) Moisture resistance Dielectric withstanding voltage Operating characteristics	Group II M8805/103-013 (4 units)		
Group III Salt spray (corrosion)	Group III M8805/103-013 (2 units)		
Group IV Low temperature operation 1/ Mechanical endurance at low temperature 2/ Mechanical endurance at high temperature 1/ Operating characteristics	Group IV M8805/103-015 (4 units)		
Group V Electrical endurance (altitude ratings only)	Group V M8805/103-013 (6 units)		

1/ Two sample units only. For low temperature operation, lever shall be held in momentary position.

2/ Same two sample units as for low temperature operation.

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(A) TABLE V. Group A inspection.

Examination or test	AQL (Percent defective)	
	Major	Minor
Visual and mechanical examination - - - -	1.0 1/	---
Operating characteristics - - - - - - - -	1.0 2/	---
Toggle to bushing resistance - - - - - -	1.0	---

1/ At the option of the manufacturer, in-process inspection may be used to verify material requirements and design and construction requirements provided they meet the acceptable quality level (AQL) as a minimum.

2/ 1.0 percent AQL applied to each applicable characteristic.

Custodian:
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

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